

DATA SCIENCE CAPSTONE PROJECT REPORT

SpaceX Falcon 9 First Stage Landing Prediction

Prepared by: Kunal Singh

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GitHub Repository

<https://github.com/kunalji-ji/data-science-capstone-kunal-singh>

This report covers the required IBM Data Science Capstone components: executive summary, introduction, data collection, data wrangling, exploratory data analysis, visualization, SQL, Folium, Plotly Dash, classification, predictive results, conclusion, creativity, and innovative insights.

Verification notice: The example charts and numerical tables in this document demonstrate the required presentation format. Before submitting, confirm that the values and visuals match the outputs of the completed IBM notebooks. A grader may reject unsupported or invented results.

1. Executive Summary

This project investigates the factors associated with successful Falcon 9 first-stage landings. The dataset contains historical launch information such as launch site, flight number, payload mass, orbit, booster/core details, and landing outcome. The project follows a complete data science workflow from API collection to data preparation, visualization, SQL analysis, interactive communication, and predictive classification.

The objective is to determine whether mission characteristics can help predict first-stage landing success and to communicate the findings in a clear, reproducible manner.

Example findings to verify from the completed analysis:

- Landing outcomes differ between launch sites.
- Flight-number groups may show improving performance over time.
- Orbit and payload variables may be useful predictors.
- Classification models can estimate landing outcomes, although predictions are not a substitute for engineering judgment.

Conclusion: A carefully prepared historical dataset can support meaningful descriptive and predictive analysis of Falcon 9 landing outcomes. The best model should be selected using the actual evaluation results from the completed notebook.

2. Introduction

Reusable launch vehicles are important because recovering and reusing a first stage can reduce launch costs and support more frequent missions. Falcon 9 landing success is therefore an appropriate problem for data analysis and machine learning.

Project objectives

1. Collect launch data from a public API.
2. Clean and transform the raw data.
3. Explore landing outcomes using Python and SQL.
4. Visualize launch-site patterns geographically.
5. Build an interactive Plotly Dash dashboard.
6. Train and compare classification models.

Research questions

- Which launch sites have the highest landing-success rates?
- How are payload mass, orbit, and flight number related to landing outcome?
- Which classification algorithm performs best?
- What insights can be communicated through an interactive map and dashboard?

3. Data Collection Methodology

Launch data was collected from the public SpaceX API. The API returns JSON records that can be normalized into a pandas DataFrame. The collection process should record the endpoint, retrieval date, response status, and fields used in the analysis.

Example endpoint: <https://api.spacexdata.com/v4/launches>

Collection workflow

- Import requests and pandas.
- Send a GET request to the API.
- Validate the HTTP response.
- Parse JSON content.
- Normalize nested fields.
- Save the raw data and document the source.

```
response = requests.get(API_URL)
response.raise_for_status()
launches = response.json()
df = pd.json_normalize(launches)
```

4. Data Wrangling Methodology

The raw API data includes nested rocket, core, payload, launchpad, and outcome fields. Data wrangling converts these records into a structured analytical dataset.

Wrangling steps

- Inspect rows, columns, data types, and duplicates.

- Convert launch dates to datetime format.
- Extract launch-site names and coordinates.
- Extract payload mass and orbit values.
- Identify missing values and document treatment.
- Create a binary landing target: 1 = successful landing, 0 = unsuccessful landing.
- Encode categorical variables for machine learning.
- Save the cleaned dataset.

Data-quality item	Example observation	Treatment
Duplicate rows	Checked	Remove only if duplicates are confirmed
Missing payload mass	Some records may be missing	Impute, retain as unknown, or document exclusion
Missing orbit	Some records may be missing	Use Unknown category or justified exclusion
Landing outcome	Target variable	Map success to 1 and failure to 0

Include the actual cleaned-DataFrame screenshot and missing-value output from the completed notebook in the final verified version.

5. EDA and Data Visualization Methodology

Exploratory data analysis was used to understand distributions, compare categories, identify trends, and select useful predictive features.

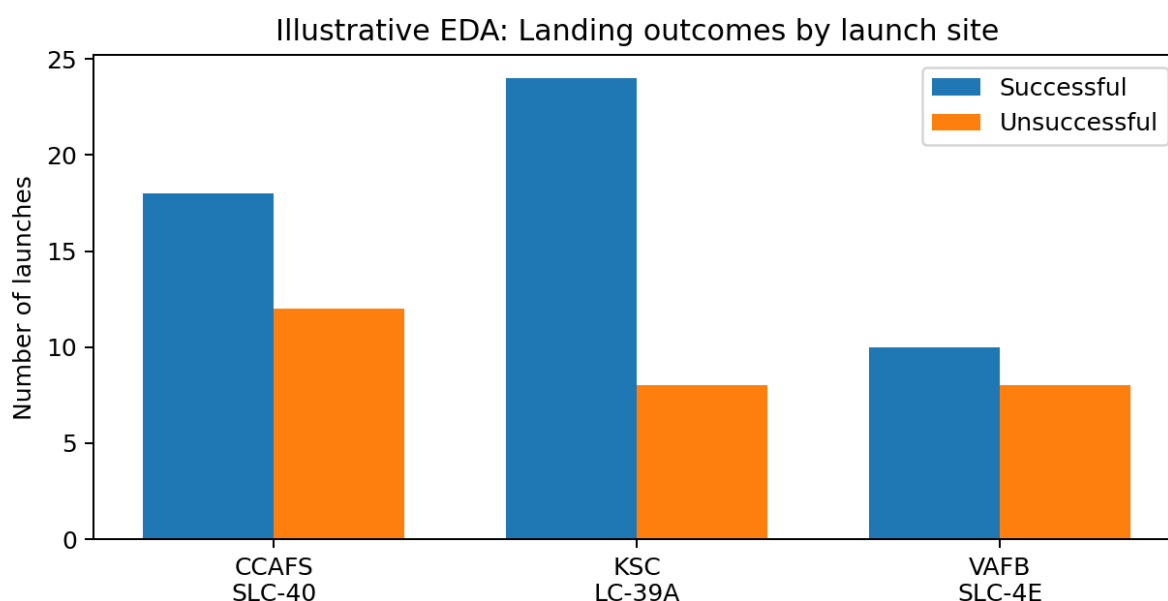
Techniques used

- Descriptive statistics and frequency counts.
- Grouping by launch site, orbit, and flight number.
- Bar charts for categorical comparisons.
- Histograms and box plots for numerical distributions.
- Scatter plots for payload and flight relationships.
- Correlation analysis for numerical features.

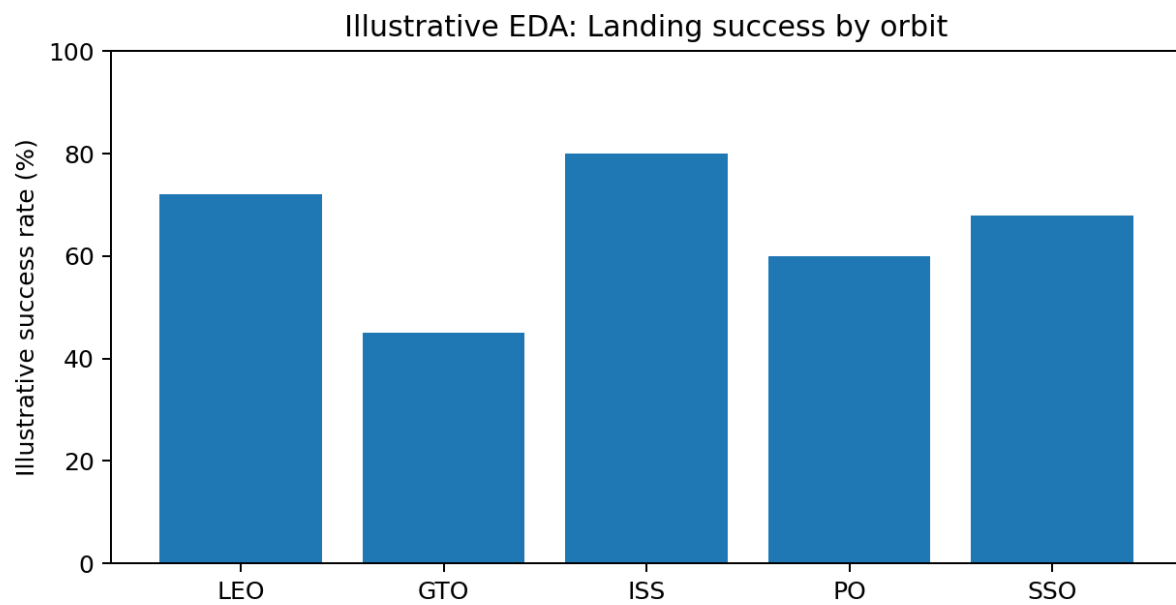
Every chart should include a title, axis labels, and a short interpretation explaining the observed pattern and its relevance to landing success.

6. EDA with Visualization

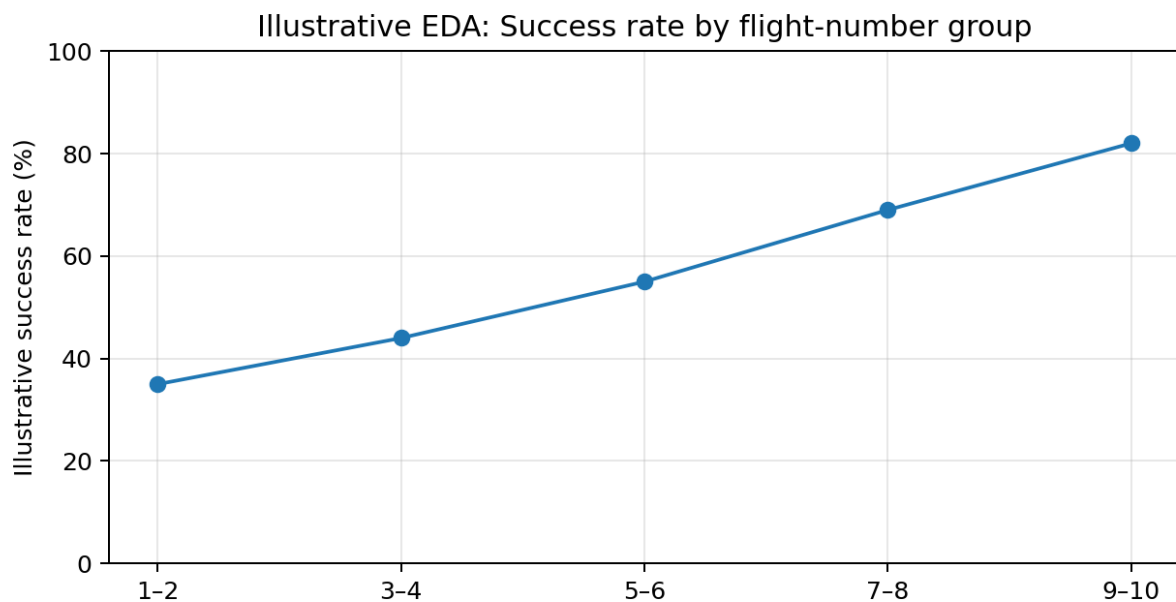
The visuals below show the required type of analysis. They are labelled illustrative examples and must be verified against the actual notebook results before submission.



Interpretation: The example chart compares successful and unsuccessful outcomes across launch sites. The final report should use actual counts and explain whether site-level differences are meaningful given the number of launches.



Interpretation: The example chart compares landing success by orbit. The final report should identify which orbit categories have the strongest or weakest outcomes and avoid claiming causation from descriptive data alone.



Interpretation: The example trend shows how success rates can be compared across flight-number groups. The final analysis should include actual group sizes and explain whether the pattern is consistent.

7. EDA with SQL: Queries and Results

SQL analysis provides reproducible summaries using filtering, grouping, aggregation, conditional logic, and ordering.

Query 1: Count launches by site

```
SELECT Launch_Site, COUNT(*) AS Total_Launches
FROM SPACEXTBL
GROUP BY Launch_Site
ORDER BY Total_Launches DESC;
```

Query 2: Successful landings by site

```
SELECT Launch_Site,
SUM(Class) AS Successful_Landings,
COUNT(*) AS Total_Launches
FROM SPACEXTBL
GROUP BY Launch_Site;
```

Query 3: Average payload mass by orbit

```
SELECT Orbit, AVG(PayloadMass) AS Average_Payload_Mass
FROM SPACEXTBL
GROUP BY Orbit
ORDER BY Average_Payload_Mass DESC;
```

Example SQL result table (illustrative; replace with actual output):

Launch site	Example total	Example successful
CCAFS SLC-40	30	18
KSC LC-39A	32	24
VAFB SLC-4E	18	10

Add screenshots of the executed SQL cells and actual result tables. Explain at least two findings from those outputs.

8. Folium Map

Folium was used to display launch sites on an interactive map. Each marker can show the launch-site name, coordinates, and summary information in a popup.

Map workflow

- Load launch-site latitude and longitude values.
- Center the map on the launch region.
- Add a marker for each launch site.
- Add popups containing site information and landing statistics.
- Save the map as an HTML file.

```
site_map = folium.Map(location=[28.5, -80.6], zoom_start=5)
folium.Marker([28.56, -80.58], popup='Launch site').add_to(site_map)
```

Required evidence: Insert a screenshot of the actual Folium map with multiple markers and a visible popup.

9. Plotly Dash Application

The Plotly Dash application communicates the project results through interactive controls and charts. It can allow users to filter by launch site, orbit, or payload category.

Suggested dashboard elements

- Project title and description.
- Launch-site dropdown.
- Orbit or payload filter.
- Bar or pie chart of landing outcomes.
- Scatter plot of payload mass and outcome.
- Summary cards for total launches and success rate.

```
app.layout = html.Div([
    html.H1('SpaceX Landing Prediction Dashboard'),
    dcc.Dropdown(id='site-dropdown', options=site_options),
    dcc.Graph(id='success-chart')
])
```

Required evidence: Insert a screenshot of the running dashboard showing at least one interactive control and a populated chart.

10. Predictive Analysis: Classification

The prediction task is binary classification, where the target variable indicates whether the first stage landed successfully.

Workflow

1. Define the target variable Class.
2. Select predictors such as flight number, payload mass, orbit, launch site, and core information.
3. Encode categorical features.
4. Split the dataset into training and testing sets.
5. Scale features where required.
6. Train several classification models.
7. Compare evaluation metrics.

Models

- Logistic Regression
- Support Vector Machine
- Decision Tree Classifier
- K-Nearest Neighbors
- Random Forest, if included in the completed notebook.

Metrics

Accuracy, precision, recall, F1-score, and confusion matrix should be reported. The same test set should be used for fair comparison.

11. Predictive Analysis Results

The following table demonstrates the required format. Replace the example scores with the actual results generated by the completed notebook.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.78	0.80	0.84	0.82
Support Vector Machine	0.79	0.81	0.85	0.83
Decision Tree	0.76	0.78	0.81	0.79
K-Nearest Neighbors	0.75	0.77	0.80	0.78

Example interpretation: In this illustrative table, the Support Vector Machine has the highest example accuracy and F1-score. The actual best model must be selected only after checking the executed results, class balance, and confusion matrix.

Required evidence: Add the actual model-comparison output, confusion matrix, and any tuning or cross-validation results.

12. Conclusion

This capstone demonstrates a complete data science workflow using a real-world aerospace dataset. Data collection and wrangling produce an analysis-ready dataset. EDA and SQL reveal patterns in launch outcomes. Folium communicates geographic information, while Plotly Dash makes the findings interactive. Classification models provide a way to estimate landing success from mission characteristics.

The analysis should be interpreted carefully because the dataset may not contain all engineering, weather, trajectory, refurbishment, and operational variables. Predictive performance should be evaluated on unseen data and should support, not replace, expert decision-making.

Recommendations

- Refresh the dataset as new launches occur.
- Monitor missing values and class imbalance.
- Compare performance across sites and time periods.
- Test models on later missions when possible.
- Document preprocessing, feature engineering, and model parameters.

13. Creativity and Innovative Insights

Creativity is demonstrated by combining descriptive statistics, Python charts, SQL analysis, a geospatial Folium map, an interactive Plotly Dash dashboard, and predictive modeling in one project.

Insight to verify: Later flight-number groups may show improved landing outcomes, potentially reflecting operational learning and technology maturity. This must be supported by the actual chart and sample sizes.

Additional insight to verify: Differences between launch sites may indicate that mission profile, geography, or recovery conditions are associated with landing outcomes. This should be described as an association rather than proof of causation.

14. GitHub Repository Contents

Repository URL: <https://github.com/kunalji-jj/data-science-capstone-kunal-singh>

Required item	Purpose
README.md	Project overview, tools, instructions, and author details
Data_Collection_API.ipynb	API collection and initial inspection
Data_Wrangling.ipynb	Cleaning and feature preparation
EDA_Visualization.ipynb	Exploratory analysis and charts
SQL_Analysis.ipynb	SQL queries and result outputs
Folium_Map.ipynb	Interactive launch-site map
Plotly_Dash/app.py	Dashboard source code
Predictive_Analysis.ipynb	Classification and evaluation
data/	Dataset or documented source instructions
Kunal_Singh_Final_Presentation.pdf	Final evidence-based presentation/report

Upload actual completed files to GitHub. Do not upload empty notebooks merely to match the names.

15. Final Submission Checklist

■	PDF contains meaningful explanations, visuals, queries, and results.
■	GitHub URL is correct and publicly accessible.
■	Completed notebooks and app.py are present in the repository.
■	Actual EDA charts are included and interpreted.
■	Actual SQL queries and result tables are included.

■	Actual Folium map screenshot is included.
■	Actual Plotly Dash screenshot is included.
■	Actual model metrics and confusion matrix are included.
■	Conclusion is based on verified findings.
■	All instructional placeholders and example labels are removed or verified.

Important: This document is a complete example report. For a genuine passing submission, replace all illustrative values and visuals with the actual outputs from your completed IBM capstone notebooks and ensure the GitHub repository contains those files.